MTH309 PRACTICE PROBLEMS WEEK 8

Question 1. Let f, g be Borel measurable, integrable functions such that

$$\int_A f\, d\mu \leq \int_A g\, d\mu,\ \forall A \in \mathcal{F}.$$

Show that $f \leq g$, μ-a.e..

Question 2. Suppose $X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$. If a.s. $|X_n| \leq Y, \forall n$ for some RV Y, then show that a.s. $|X| \leq Y$.

Question 3. Let $X, Y : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathbb{B}_{\mathbb{R}})$ be two RVs such that $\mathbb{E}|X - Y| = 0$. Show that $X = Y$ a.s..

Question 4. Let X be an RV defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose $Y : \Omega \to \mathbb{R}$ be a function such that $X = Y$ a.s. (almost surely). Is Y an RV?

Question 5. Let $\{f_n\}_n$ be a sequence of real valued measurable functions on a measure space $(\Omega, \mathcal{F}, \mu)$ such that

$$\sup_n \int_\Omega |f_n|^{1+\epsilon}\, d\mu < \infty$$

for some $\epsilon > 0$. Show that the sequence is uniformly integrable, i.e. the following holds

$$\lim_{x\to\infty} \sup_n \int_{(|f_n|>x)} |f_n|\, d\mu = 0.$$

Question 6. Fix $a, b \in \mathbb{R}$ with $a < b$. Suppose that there exist functions $f : (a, b) \times \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ such that

(a) $\omega \in \Omega \mapsto f(t, \omega)$ is μ-integrable for every fixed $t \in (a, b)$,

(b) $t \in (a, b) \mapsto f(t, \omega)$ is continuous for every fixed $\omega \in \Omega$,

(c) $g \geq 0$ and μ-integrable,

(d) $|f(t, \omega)| \leq g(\omega), \forall t, \omega$.

Then show that the function $h : (a, b) \to \mathbb{R}$ defined by

$$h(t) := \int_\Omega f(t, \omega)\, d\mu(\omega)$$

is continuous.

Question 7. Fix $a, b \in \mathbb{R}$ with $a < b$. Suppose that there exist functions $f : (a, b) \times \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ such that

(a) $\omega \in \Omega \mapsto f(t, \omega)$ is μ-integrable for every fixed $t \in (a, b)$,
(b) $t \in (a, b) \mapsto f(t, \omega)$ is differentiable (in t) for every fixed $\omega \in \Omega$,
(c) $g \geq 0$ and μ-integrable,
(d) $|\frac{\partial}{\partial t} f(t, \omega)| \leq g(\omega), \forall t, \omega$.

Then show that the function $h : (a, b) \to \mathbb{R}$ defined by

$$h(t) := \int_\Omega f(t, \omega)\, d\mu(\omega)$$

is differentiable with $h'(t) = \int_\Omega \frac{\partial}{\partial t} f(t, \omega)\, d\mu(\omega)$.

Question 8. Prove the folllowing statement left out in the class. Let $f : (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$ and $g : (\Omega_2, \mathcal{F}_2) \to (\bar{\mathbb{R}}, \mathbb{B}_{\bar{\mathbb{R}}})$ be measurable functions. Suppose μ_1 is a measure on $(\Omega_1, \mathcal{F}_1)$ and define $\mu_2 := \mu_1 \circ f^{-1}$ (the push-forward measure). Show that

$$\int_{f^{-1}(A)} g \circ f(\omega_1)\, d\mu_1(\omega_1) = \int_A g(\omega_2)\, d\mu_2(\omega_2),\ \forall A \in \mathcal{F}_2.$$

Question 9. (AshBk p. 55) Show that

$$\int_0^1 e^{-t} \ln t\, dt = \lim_{n \to \infty} \int_0^1 \left[1 - \frac{t}{n}\right]^n \ln t\, dt$$

Question 10. Let $(\Omega, \mathcal{F})$ be a measurable space. Suppose $\mathbb{P}_1$ and $\mathbb{P}_2$ are two probability measures on this space. Show that that the following conditions are equivalent.

(a) For any $A \in \mathcal{F}$, $\mathbb{P}_1(A) = 0$ implies $\mathbb{P}_2(A) = 0$.
(b) For every $\epsilon > 0$, there exists $\delta > 0$ such that if $A \in \mathcal{F}$ with $\mathbb{P}_1(A) < \delta$, then $\mathbb{P}_2(A) < \epsilon$.

Question 11. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Consider the following two collections:

$$\mathcal{C} = \{A \in \mathcal{F} : \mathbb{P}(A) = 0\}, \quad \mathcal{D} = \{A \in \mathcal{F} : \mathbb{P}(A) = 1\}.$$

Check if the following properties are true.

(a) Are the collections non-empty?
(b) Are the collections closed under complementation?
(c) Are the collections closed under finite unions?
(d) Are the collections closed under countable unions?
(e) Are the collections closed under finite intersections?
(f) Are the collections closed under countable intersections?

(g) Are the collections closed under countable increasing unions?

(h) Are the collections closed under countable decreasing intersections?